**Development of Python Code Compatible with Multiple AI Tools**

**Aim**

To develop a Python program that integrates multiple AI tools using APIs, compares their outputs, and generates actionable insights.

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**Explanation (Scenario-Based)**

In this experiment, we simulate interaction with multiple AI tools such as:

- OpenAI (ChatGPT)
- Other AI APIs (e.g., Cohere, Hugging Face, etc.)

The system will:

1. Send the same prompt to multiple AI APIs
2. Collect responses
3. Compare outputs
4. Generate insights like:
    - Which response is longer?
    - Which is more detailed?
    - Similarity between responses

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**Python Implementation**

```
import requests

# -------------------------------
# Step 1: Define API Keys
# -------------------------------
OPENAI_API_KEY = "your_openai_api_key"
HUGGINGFACE_API_KEY = "your_huggingface_api_key"

# -------------------------------
# Step 2: Function for OpenAI API
# -------------------------------
def get_openai_response(prompt):
    url = "https://api.openai.com/v1/chat/completions"

    headers = {
        "Authorization": f"Bearer {OPENAI_API_KEY}",
        "Content-Type": "application/json"
    }

    data = {
        "model": "gpt-4o-mini",
```

```
        "messages": [{"role": "user", "content": prompt}]
    }

    response = requests.post(url, headers=headers, json=data)
    result = response.json()

    return result['choices'][0]['message']['content']

# -------------------------------
# Step 3: Function for HuggingFace API
# -------------------------------
def get_huggingface_response(prompt):
    url = "https://api-inference.huggingface.co/models/gpt2"

    headers = {
        "Authorization": f"Bearer {HUGGINGFACE_API_KEY}"
    }

    data = {"inputs": prompt}

    response = requests.post(url, headers=headers, json=data)
    result = response.json()

    return result[0]['generated_text']

# -------------------------------
# Step 4: Compare Outputs
# -------------------------------
def compare_outputs(resp1, resp2):
    insights = {}

    insights['OpenAI_length'] = len(resp1)
    insights['HuggingFace_length'] = len(resp2)

    if len(resp1) > len(resp2):
        insights['Detailed_Response'] = "OpenAI"
    else:
        insights['Detailed_Response'] = "HuggingFace"

    common_words = set(resp1.split()).intersection(set(resp2.split()))
    insights['Common_Words_Count'] = len(common_words)

    return insights

# -------------------------------
# Step 5: Main Execution
# -------------------------------
if __name__ == "__main__":
    prompt = "Explain Artificial Intelligence in simple terms"

    print("Sending request to OpenAI...")
```

```
    openai_output = get_openai_response(prompt)

    print("Sending request to HuggingFace...")
    hf_output = get_huggingface_response(prompt)

    print("\n--- OpenAI Response ---")
    print(openai_output)

    print("\n--- HuggingFace Response ---")
    print(hf_output)

    # Compare results
    insights = compare_outputs(openai_output, hf_output)

    print("\n--- Insights ---")
    for key, value in insights.items():
        print(f"{key}: {value}")
```

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**Output Analysis**

After running the program, you will get:

**1. Responses from both AI tools**

- OpenAI → usually structured, accurate
- Hugging Face → may be raw or less structured

**2. Comparison Insights**

Example:

OpenAI_length: 320
HuggingFace_length: 180
Detailed_Response: OpenAI
Common_Words_Count: 25

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**Result**

- Successfully integrated multiple AI APIs
- Compared outputs automatically
- Generated useful insights from responses

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**Conclusion**

This experiment demonstrates:

- How to interact with multiple AI tools using APIs
- How to automate response comparison

- How to extract meaningful insights